

# IMD Doppler Weather Radar (DWR) Operation & Dual-Polarization Guide

Operational Handbook on S-band/C-band Hydrometeor Classification (ZDR, KDP, CC), Nyquist De-aliasing, and Severe Cyclone Nowcasting

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## 1. Foundations of Dual-Polarization Radar Meteorology

Dual-Polarization Doppler Weather Radars transmit and receive radio microwave pulses in both horizontal and vertical polarization orientations. By computing phase and amplitude differences between orthogonal channels, forecasters can unambiguously determine hydrometeor size, shape, phase state (liquid, ice, mixed), and spatial orientation in real-time convective storm cores.

| Polarimetric Variable              | Mathematical Definition                      | Typical Range     | Physical Interpretation                       |
|------------------------------------|----------------------------------------------|-------------------|-----------------------------------------------|
| Reflectivity (Z)                   | $Z = \int N(D) D^6 dD$                       | -10 to +75 dBZ    | Precipitation intensity and droplet density   |
| Diff. Reflectivity ( $Z_{DR}$ )    | $10 \log_{10}(Z_H / Z_V)$                    | -1.0 to +5.0 dB   | Oblateness of raindrops vs tumbling hail      |
| Correlation Coeff. ( $\rho_{HV}$ ) | Cross-channel correlation                    | 0.70 to 0.99      | Hydrometeor homogeneity; drops <0.90 for hail |
| Specific Diff. Phase ( $K_{DP}$ )  | $\partial(\Phi_{DP}) / \partial r \cdot 0.5$ | -0.5 to +8.0 °/km | Liquid water content, immune to attenuation   |

## 2. Hydrometeor Classification Algorithm (HCA) Matrix

Operational Doppler Radars across coastal and inland IMD stations deploy fuzzy-logic HCA to classify targets into 10 meteorological and non-meteorological classes within seconds of volume scan completion.

**Severe Hail Identification Rule:** High horizontal reflectivity ( $Z_H > 55$  dBZ) accompanied by near-zero Differential Reflectivity ( $-0.2 \leq Z_{DR} \leq 0.5$  dB) and low Correlation Coefficient ( $\rho_{HV} < 0.92$ ) is the definitive signature of large, tumbling, spheroidal hail aloft.

## 3. Doppler Velocity De-aliasing & Nyquist Interval

The Maximum Unambiguous Velocity is bounded by the pulse repetition frequency:  $V_{max} = (\lambda \cdot PRF) / 4$ . Operational dual-PRF algorithms (e.g. 3:2 or 4:3 ratios) extend unambiguous velocity up to 64 m/s to capture hurricane-force eyewall rotation during intense Arabian Sea and Bay of Bengal tropical cyclones.